

Professional commitment and health among teachers in Germany: A typological approach

Ulf Kieschke*, Uwe Schaarschmidt

Department of Psychology, University of Potsdam, PF 601553, 14415 Potsdam, Germany

Abstract

The interplay of occupational engagement and individual coping capacity is very important. A methodological tool that provides the means for such an analysis which includes psychological aspects of the health is the “Arbeitsbezogenes Verhaltens- und Erlebensmuster” questionnaire (AVEM: Pattern of Work-related Coping Behavior). This assessment instrument allows for a subscale-level evaluation as well as differentiation between four types of work-related coping behaviours that have been obtained by cluster analysis. Comprehensive analyses involving 7693 teachers have established the health relevance of this cluster differentiation.
© 2008 Elsevier Ltd. All rights reserved.

Keywords: Professional commitment; Personality; Coping styles; Occupational stress

1. Introduction

Teaching is a challenging and, therefore, often stressful job. Undoubtedly, teachers face great demands on their energy and personal skills (Montgomery & Rupp, 2005; Schaarschmidt, 2005; Vandenberghe & Huberman, 1999). Sometimes this is more than a person with few coping resources can handle. In Germany, the situation has become critical over the last years, with alarming numbers of teachers taking disability leave or early retirement as a result of stress-related conditions (Weber, 2003, 2004). According to statistics from 2001, 72% of German teachers left the profession before reaching the regular retirement age (compared with 41% for other officials in the public service; Bundesinnenministerium, 2001). Very often in Germany, and more than in other occupations, teachers are treated for psychosomatic conditions (Hillert & Schmitz, 2004), with teachers complaining about a constant increase in their workload coupled with declining working conditions, particularly from problems with students’ behaviours and decreasing assistance in schools. It is therefore essential to enhance psychological health among teachers.

At the University of Potsdam we have been studying teachers’ psychological health in German schools for 10 years. Especially, we have investigated personal resources and competences which facilitate health-supporting management for their challenges at work. To highlight the stressful situation of teachers we have included comparative analyses about work-related coping behaviours among other occupations. The basis for these comparisons has been data that we collected with the “Arbeitsbezogenes Verhaltens- und Erlebensmuster” questionnaire

* Corresponding author.

E-mail address: kieschke@rz.uni-potsdam.de (U. Kieschke).

(AVEM: Pattern of Work-related Coping Behavior; Schaarschmidt & Fischer, 2003). More than 7600 teachers as well as 6000 professionals from other occupations have taken part in our studies. Meanwhile longitudinal studies have taken place with a teacher sample ($N = 291$).

This article highlights key findings from our research among teachers in Germany. The question of motivation — the focus of this special issue — is a central concern within our approach and conclusions. As we emphasize, strong professional commitment is not sufficient to find satisfaction and happiness in working life. If teachers carry out their duties too much and expect too much of themselves and their surroundings they run the risk of experiencing heightened stress and disappointments (Jepson & Forrest, 2006). That in turn will have long-term negative consequences for their teaching effectiveness and classroom management (Klusmann, Kunter, Trautwein, & Baumert, 2006). The quality of academic learning opportunities can only be enhanced by competent, committed *and* healthy teachers (Prenzel et al., 2005).

2. Work, personality and health: the AVEM typology

For decades, psychometric instruments have been used to measure the detrimental effects of work on people's health (Hart & Cooper, 2002). The employment of numerous symptom and problem checklists reflects the central role that is ascribed to the psychological and somatic indicators of strain. The conceptualization of the burnout syndrome is also dominated by its physical or psychological effects (Vandenberghe & Huberman, 1999). Based on modern models of health psychology, current research has gone beyond the mere identification of problems and disorders. For example, the salutogenetic approach of Antonovsky (1987) suggests taking into account personal and social resources as supporting and buffering factors. Here, the affected person is perceived as an active element in dealing with professional demands and strains rather than a passive victim of the circumstances. Analyzing one's coping resources can lead to the early identification of possible future health risks. This constitutes an important advantage: recommendation for intervention can be given on the basis of identified deficits in coping, and can be applied before symptoms have occurred. In fact, utilizing one's resources is far more effective than intervention after disorders have become manifest. The AVEM questionnaire (Schaarschmidt & Fischer, 2003) allows a detailed assessment of an individual's developmental potential with special regard to aspects of psychological health. This instrument has been designed to gather self-report data about personal experiences with work-related stress, and typical behavioral responses in coping with stress.

In detail, the questionnaire covers the following three major areas: professional commitment, coping capacity, and subjective well-being (in the context of work) which are assessed with 11 separate scales (see Fig. 1 for details). Each scale contains 6 items with responses in a Likert-type format ranging from 1 (strongly disagree) to 5 (strongly agree).

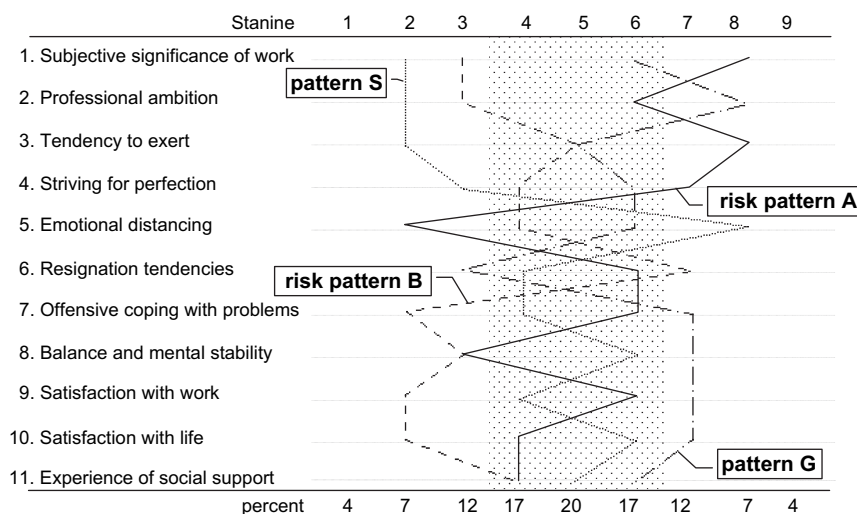


Fig. 1. Four patterns of work-related coping behavior: G, S, A, B. The 11 scales can be divided into three areas of characteristics: professional commitment (scales 1–5), coping capacity (scales 6–8), and subjective well-being (scales 9–11).

In order to determine an individual's pattern of behavioral responses to occupational stress it is necessary to go beyond the analysis of singular scales. Information on the configuration of characteristics across the dimensions is diagnostically more valuable than summary scores for the separate dimensions. For example, high professional commitment in itself does not constitute a health risk. If, however, a tendency to high professional commitment is coupled with an impaired coping capacity, then this combination of two traits may set the individual at a higher risk for developing health problems in the future (see Houston, Chesney, Black, Cates, & Hecker, 1992).

To identify characteristic configurations of behavioral traits that may allow assumptions about a person's health status and risk, Schaarschmidt and Fischer (2003) explored the data from the initial AVEM sample ($N = 1598$), containing professionals from diverse occupations. Cluster analysis yielded a four-cluster solution (Fig. 1). The four-cluster model was conceptually plausible, satisfied the formal quality standards for multivariate classification techniques (i.e., high within-cluster homogeneity and high between-cluster heterogeneity), and has been shown to fulfill the psychometric prerequisites for extensive profile analyses (i.e., high reliability and moderate intercorrelation of singular scales). Moreover, the same cluster solution was replicated with sufficient concurrence in 10 random samples out of the total group of 1598 test participants (average $\kappa > 0.80$). The four clusters are the following:

2.1. Pattern/type 'G'

The letter 'G' stands for *good health*. Individuals who display this pattern are in good psychological health and show a health-promoting attitude towards their work. G-type individuals score high but not highest on professional commitment (including the subjective significance of work, professional ambitions, tendency to exert, striving for perfection). Although the will to show high performance is expressed, the ability to distance oneself from work-related demands is preserved (emotional distancing). Those in this cluster are good at relaxing; they are able to prevent intrusion of professional problems into their leisure time. Scores on the scales measuring coping capacity further underline their healthy attitude towards work (low scores on resignation tendencies, high scores in offensive coping with problems, balance and mental stability). The profile of the G-type person is completed with high scores on those scales which indicate a generally positive attitude towards life (satisfaction with work, satisfaction with life, experience of social support). The G-pattern in and of itself does not guarantee a good teacher, but it can be assumed that this profile possesses the optimal conditions to actualize their knowledge and competencies, pedagogical beliefs, and intentions.

2.2. Pattern/type 'S'

The letter 'S' marks *an attitude of sparing personal investment at work*. People belonging to this cluster restrict their efforts at work to what is absolutely necessary. Characteristics of the S-type include below average scores on subjective significance of work, professional ambition, tendency to exert, and striving for perfection. In addition, individuals who display this behavioral pattern score the highest of all the four types on emotional distancing. The tendency towards resignation is rather low which further indicates that low commitment in the work environment is not tantamount to despair and resignation. Instead, the low tendency towards resignation is expressed in conjunction with high scores on the scale for balance and mental stability. This combination suggests a rather effective coping capacity when professional challenges are faced. Finally, even though the S-type individual's positive attitude towards life (expressed by relatively high life satisfaction scores) may be health protective, it is probably not primarily derived from success at work, as indicated by low scores on perceptions of professional success.

In general, the S-pattern may be of more interest in terms of their professional motivation than occupational health. Although they possess positive psychological health, perhaps their low commitment to work is not what we might desire, especially among school teachers. Nevertheless, as for all the types, it should be noted that behaviours as described by this pattern are not solely an expression of personality traits. In the case of the S-type for example, a lack of professional challenges may have contributed to their adoption of this attitude towards work. Moreover, the display of an "energy conserving" attitude may be beneficial for health (e.g., protection against excessive demands when work conditions are poor).

2.3. Risk pattern/type 'A'

The letter 'A' stands for *ambitious*. This pattern is characterized by excessive commitment at the workplace. The scores on relevant scales show significant deviations from average, such as their low ability to distance themselves from work. Further, high commitment is not coupled with sufficient resilience (low scores on coping capacity) to withstand excessive stress and work-related demands which set the individual at a high risk for developing health problems. In addition, quite negative emotions dominate the work experience. High commitment to work does not seem to be rewarding, at least not in terms of an "emotional benefit". The discrepancy between professional commitment and the absence of success has been described by Siegrist (1991) as "gratification crisis". In other words, high professional commitment is insufficiently appreciated and leads to low satisfaction. In the long run, this leads to increased health problems such as cardiovascular diseases (Siegrist, 1991). Type A individuals show many similarities with the well-known type A behavior concept introduced several decades ago by Friedman and Rosenman (1974). The pattern we describe attempts to broaden their original concept by emphasizing the complex interplay of excessive commitment, lowered coping capacity, and negative emotions for an individual's health.

2.4. Risk pattern/type 'B'

The letter 'B' stands for *burnout*. Representatives of pattern 'B' reflecting *burnout* are characterized by low scores in the dimensions of professional commitment, especially in the subjective significance of work and professional ambitions. Similarities with the S-type are evident, yet there are several pivotal differences. In contrast to the S-type, reduced commitment among type B is not connected to a higher, but rather to a limited ability to distance oneself from work demands. For the coping capacity dimensions, all scores are problematic (i.e., resignation tendencies, offensive coping with problems, balance and mental stability). Moreover, low scores on virtually all scales related to satisfaction with work and well-being indicate a negative emotional tone. This type is characterized by exhaustion, the experience of excessive challenge, and resignation. Similarities to the burnout syndrome described by Freudenberg (1974) and Maslach (1982) are obvious. Referring to the pattern as 'B' (*burnout*) emphasizes this parallel.

These four experiential and behavioral types are the basis of our classification according to the AVEM instrument. Each individual is assigned to one of the four types (G, S, A, or B) based on a weighted linear combination of AVEM scale scores (for the technological—statistical procedure based on an algorithm of discriminant analysis, see Kieschke, 2003; Schaarschmidt & Fischer, 2003). Individuals are classified as belonging to the type most closely matching their individual profile of scores. For example, if a person achieves the probability scores $p_G = 0.70$, $p_S = 0.20$, $p_A = 0.06$, and $p_B = 0.04$, he/she is classified as type G — the type to which s/he has the highest probability of belonging. The AVEM classifications are therefore statements about the approximate similarities between individual cases and their group memberships. We have demonstrated in a number of studies that the AVEM is a reliable and easy to apply instrument when used to differentiate work-related coping efforts (Kieschke, 2003; Schaarschmidt & Fischer, 2001). The four coping behavior types correlate with number of sick days per year, desire for early retirement, evaluation of working conditions (measured by the extent to which they contribute to health and personal growth), physical indicators of responses to both strain and relaxation, and mental and physical problems (for details see Heitzmann, Schaarschmidt, & Kieschke, 2005; Kieschke, 2003; Kieschke & Schaarschmidt, 2003; Schaarschmidt, 2005; Schaarschmidt & Fischer, 2001, 2003). Pattern G shows the most favorable relationships, and risk patterns A and B the most unfavorable relationships with employees' health. There are important qualitative and quantitative differences between the two risk types A and B. While both are characterized by high physical disturbances (headache, neck and backache, cardiovascular disease, digestive problems), type B individuals exhibit higher levels of psychological complaints (exhaustion, decreased self-esteem, experiences of insufficient achievement) than type A (Schaarschmidt, & Fischer, 2001).

3. Selected results

The AVEM data were collected from several German samples between 2000 and 2005 in research projects on the interplay of personality and health in employment contexts (for details see Kieschke, 2003; Schaarschmidt, 2005). The following professional groups are represented in our analysis: teachers ($N = 7639$; 65.7% female; age:

$M = 47.4$ years, $SD = 9.0$); entrepreneurs and business starters ($N = 734$; 29.1% female, age: $M = 38.2$ years, $SD = 7.3$); police ($N = 851$; 18.5% female; age: $M = 41.6$ years, $SD = 8.7$); officials in penal institutions ($N = 3653$; 18.6% female; age: $M = 45.6$ years, $SD = 8.9$); firefighters ($N = 382$; 2.4% female; age: $M = 45.0$ years, $SD = 7.7$); and nurses ($N = 378$; 92.5% female; age: $M = 36.5$ years, $SD = 10.8$). Reliability statistics for the six professional groups examined are presented in Table 1.

The psychometric quality of the diagnostic tool is demonstrated both with regard to the homogeneity of the internal consistencies (Cronbach's α) within each sample, as well as similarity in those coefficients across different samples. Low correlations of the 11 AVEM scales in all six samples are a further necessary condition for classifying profiles reliably. The average value calculated with Fisher's Z aggregation was below $|0.30|$ in every sample. A comparison of the AVEM patterns across the six different professions is illustrated in Fig. 2.

Among teachers, the frequency of unfavorable patterns is disproportionately high, with almost 60% of the sample belonging to one of the two risk types A or B. In contrast, very different results emerge for the sample of entrepreneurs and business starters. Among entrepreneurs, pattern G (good health) is highly prevalent, although the percentage of risk group A is also noteworthy, and tendencies of overexertion and self-exploitation were in fact more prominent in this group than in any other (Kieschke & Schaarschmidt, 2003, 2005). The highest frequency for pattern S (spare personal investment at work) is observed among members of professional fire brigade and nurses, which may be an indication of limited work satisfaction within these professions (Schaarschmidt & Fischer, 2001). Nevertheless, the distribution of AVEM types is not only due to the specifics of the job and its working conditions. It is important to note that personality traits and motivational factors also determine individuals' career decisions (Feldman, 2002; Richardson & Watt, 2005, 2006; Watt & Richardson, 2007). Therefore, effects of self-selection have to be taken into account in interpreting these results (e.g., Lent, Brown, & Hackett, 1994). For example, people who are shy or restrained and have no pleasure in contact with children and adolescents will be unlikely to consider becoming a teacher.

Our findings for teachers were similar across the different regions of Germany, and for teachers of different school grades, thus painting a general profile for teachers across the country. However, female teachers showed persistently significantly higher rates of risk patterns A (ambitious) and B (burnout) than males. Age trends indicated a progressive worsening of teachers' health situation according to length of time in the profession, and again this tendency was more pronounced for women than for men.

Among teacher education students, the percentage of risk type B (burnout) is already high (25%), implying the existence of unpropitious conditions before entry to the teaching profession (Schaarschmidt, 2005). However, it would be an over-simplification to attribute the problematic health situation of teachers solely to this disadvantageous precondition. There are also definite connections with time spent in the profession, and particular aspects of professional demand. The most stressful conditions reported include the behavior of difficult pupils, large class sizes, and high numbers of classes to be taught (Schaarschmidt, 2005). Even teachers with the optimal pattern G declare high stress from these factors, particularly pupil behavior and large classes; however, they additionally report relieving conditions, most importantly the experience of social support. Supportive school administration and colleagues were

Table 1
Reliability statistics of the AVEM subscales (internal consistencies) in a cross-professional comparison

Subscale (sample item)	TEA	OPI	PPE	PFB	NUR	EPR
1. Subjective significance of work (sample item: "Work is the most important element in my life.")	0.84	0.84	0.85	0.86	0.83	0.84
2. Professional ambition (sample item: "I want to achieve more in my career than most people I know.")	0.83	0.85	0.86	0.85	0.84	0.81
3. Tendency to exert (sample item: "If necessary, I will work until I am exhausted.")	0.83	0.79	0.74	0.75	0.76	0.79
4. Striving for perfection (Sample item: "My work should never contain errors or deficiencies.")	0.86	0.81	0.80	0.80	0.78	0.85
5. Emotional distancing (sample item: "After work is over I can forget about it quickly.")	0.86	0.85	0.86	0.83	0.87	0.85
6. Resignation tendencies (Sample item: "I quickly resign myself to lack of success.")	0.84	0.80	0.80	0.82	0.77	0.85
7. Offensive coping with problems (sample item: "For me, difficulties are there to overcome.")	0.81	0.80	0.80	0.79	0.77	0.81
8. Balance and mental stability (Sample item: "I don't get upset easily.")	0.83	0.78	0.79	0.81	0.80	0.80
9. Satisfaction with work (sample item: "Until now I have been successful in my work.")	0.85	0.82	0.83	0.81	0.78	0.85
10. Satisfaction with life (sample item: "So far, I have been satisfied with my life.")	0.83	0.75	0.78	0.81	0.80	0.79
11. Experience of social support (sample item: "My partner shows understanding for my work.")	0.82	0.74	0.74	0.75	0.75	0.81

TEA; teachers; OPI: officials in penal institutions; PPE: police personnel; PFB: professional fire brigade; NUR: nurses; EPR: entrepreneurs.

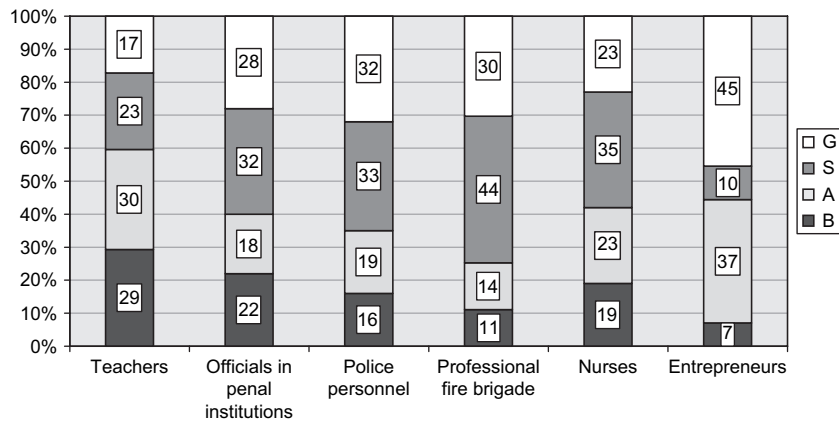


Fig. 2. Distribution of AVEM types in a cross-professional comparison.

perceived as reducing the number of days of sick leave, and also the impact of stressful conditions (Schaarschmidt & Kieschke, 2007).

Longitudinal patterns of teachers' coping were of special interest (Kieschke, 2003, 2005). Based on a three-year follow up of 291 teachers in the original sample, statements about changes in their psychological health can be made. Data were checked for systematic drop-out effects with the finding that self-selection effects, which would render the interpretation of the longitudinal findings difficult, can be largely excluded. The re-participation ratio varied neither as a function of sex, age, or pattern classification at the first measurement point.

Fig. 3 displays patterns at both measurement points, as well as crossovers between types over time. In general, changes from the first to the second occasion were detrimental. Although the decrease in type A is positive, there is an increase of the risk type B, and the frequency of the desirable type G also declines. The accretion of type S is not favorable, with nearly a quarter shifting to the burnout type B. Therefore, the initially problematic constellation of teachers' coping patterns does not seem to improve over time, implying that in the absence of intervention, the situation would continue to worsen.

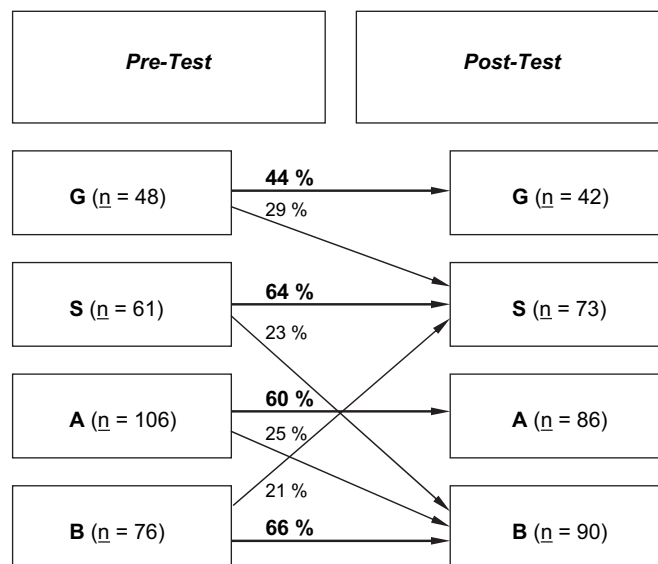


Fig. 3. Crossovers of the patterns from the first to the second time of measurement. Percentages for only the most important pattern shifts are presented, therefore do not sum to 100%. The illustration is to be read as follows: type G: 44% (of the 48 persons who have been allocated to this pattern at the first time of measurement) stay type G, 29% change to S; type S: 64% persist in S, 23% change to B; type A: 60% continue A, 25% change to B; type B: 66% remain in B, 21% change to S.

We wish to particularly highlight movement towards the problematic pattern B. As expected there was no change from type G to type B. The strongest trend into the direction of type B comes from type A: 25% of the people belonging to type A at the first time of measurement transitioned into type B, illustrating a path from excessive engagement to resignation, which can definitely be spoken of a burnout process. Many people (23%) also moved from type S to type B. This does not fit a burnout process, but rather demonstrates that when people are acting on minimal resources this can become a health risk, particularly if these persons experience criticism and are deprived of social support. In short, not all people showing pattern B have displayed a burnout process. On the one hand there are people who have changed from protection to resignation — including those teachers who started out with excessive demands and failure anxiety; and also the large numbers who stayed in the risk type B, intensified by disappointment. Consequently, those people exhibiting negative tendencies from the outset contribute to the high stability of the type B.

4. Conclusions

A considerable proportion of teachers show health risks (Friedman, 2006; Kyriacou, 2001; Lambert & McCarthy, 2006; Schaarschmidt, 2005; Schaarschmidt & Kieschke, 2007). Of course, being a teacher is not “pathological” in itself. Last but not least the personal style of coping with the demands of the occupation is essential (Richardson & Watt, 2005; Schaarschmidt, 2005). The introduced type system based upon the AVEM (Schaarschmidt & Fischer, 2003) provides a theoretical and methodological framework for the description of such personal styles. Four distinct patterns of coping with professional demands have been identified and labelled: G, S, A, and B. The AVEM approach is conceptually plausible, it meets the formal quality standards for psychological assessment, and has shown to be a valid system of empirical comparison (Kieschke, 2003; Schaarschmidt, 2005; Schaarschmidt & Kieschke, 2007). It has proven to be a useful research instrument as well as a diagnostic tool for individual examinations.

It is critical that the two risk patterns A and B predominate among German teachers. Representatives of pattern A are characterized by excessive commitment coupled with inadequate coping resources; and representatives of pattern B are too exhausted to be capable of big efforts. While type A seems to be highly engaged in their work, we see symptoms of burnout among B representatives (Schaarschmidt & Fischer, 2003). Both patterns are highly problematic — on the one hand in relation to their psychological health, and on the other regarding their quality of instruction and education. In fact, recent studies have shown that the type differentiation among teachers is reflected in students’ reports of their instructional behavior (Klusmann et al., 2006). Among the four types of the AVEM, G-type teachers came off best. G-representatives are most able to support learning processes constructively and strengthen the cognitive independence of pupils (Baumert & Kunter, 2006; Klusmann et al., 2006). Against this background the reported longitudinal results deserve special attention. The risk pattern quota do not reduce spontaneously. The problematic pattern B also proved to be the most stable profile in terms of health and occupational competence. Further, we observed a considerable number of changes from A to B, a transition that can be described as a “burnout” process (Schaarschmidt, 2005; Vandenberghe & Huberman, 1999).

Keeping in mind that the longitudinal investigation only spanned a time interval of three years, it is likely that if the interval between pre- and posttest were increased, there may be still more teachers experiencing this burnout process, since the three-year interval may underestimate the long-term effects of work stress. A focus on differential developmental processes would be a profitable direction for future research. Which organizational circumstances and personal factors enhance the probability of drifting into pattern B? Some indications are suggested in our current studies (see Schaarschmidt & Kieschke, 2007). Deficits in school administration and poor social cohesion are associated with unfavorable pattern distribution (Laux, Ksienzyk-Kreuziger, & Kieschke, 2007).

The personality and health-supporting conditions in our teacher sample show other points where interventions might begin to redress these problematic patterns. There are four tasks we think of as very important (Schaarschmidt & Kieschke, 2007). First, it is essential to establish the preconditions for more autonomy in teachers’ work. Many teachers have lost the joy in their occupation because too much regimentation and external interference complicates their pedagogic targets and self-determined professional goals (Evertson & Weinstein, 2006). Another result of our studies further illustrates this point: 22 work-related motives were rated by teachers in terms of their personal significance, out of which the motive to *act autonomously* was consistently named strongest (Schaarschmidt, 2005). Second, the excessive demand on teachers from overwhelming educational tasks should be avoided: they should not be left alone with the more complex and difficult professional demands. Third, necessary recovery and regeneration processes should be better facilitated. Timetabling of school lessons should allow for appreciable phases of relaxation,

rather than concentrations during blocks of time which demand sustained highest levels of attention and effort. There is also a need for a clearer separation of life at school and leisure time. The strong interdependence of both areas is due to the displacement of many school tasks into the evenings and weekends, hindering the emotional distancing between everyday life at school and opportunities for regeneration, especially for women. Fourth, it is urgent that we focus on sufficient preparation of teachers. Given the stress of the teaching occupation, it is important that positive preconditions are established prior to professional entry. We think the basic conditions for teachers are joy in associating with children and adolescents, social-communicative strengths, emotional stability, as well as a proactive approach towards life demands. It is therefore essential to directly convey effective coping skills for everyday occupational problems during teacher education; especially the capability for effective self management in stressful situations (Schaarschmidt & Kieschke, 2007). Together these approaches should make it possible to effectively promote the development of health as well as competence in the teaching profession.

References

- Antonovsky, A. (1987). *Unraveling the mystery of health: How people manage stress and stay well*. San Francisco, CA: Jossey Bass.
- Baumert, J., & Kunter, M. (2006). Stichwort: Professionelle Kompetenz von Lehrkräften. [Key word: professionalism of teachers]. *Zeitschrift für Erziehungswissenschaft*, 9, 469–520.
- Bundesinnenministerium. (2001). *Versorgungsbericht der Bundesregierung [Report about the public service in Germany]*. <http://www.bmi.bund.de/cln_028/nn_164578/Internet/Content/Broschueren/2001/Zweiter_Versorgungsbericht_der_Id_56613_de.html>. Accessed 21.06.07.
- Evertson, C. M., & Weinstein, C. S. (Eds.). (2006). *Handbook of classroom management. Research, practice, and contemporary issues*. Mahwah, NJ: Erlbaum.
- Feldman, D. C. (Ed.). (2002). *Work careers: A developmental perspective*. San Francisco, CA: Jossey Bass.
- Freudenberger, H. J. (1974). Staff burn-out. *Journal of Social Issues*, 30, 159–165.
- Friedman, I. A. (2006). Classroom management and teacher stress and burnout. In C. M. Evertson, & C. S. Weinstein (Eds.), *Handbook of classroom management: Research, practice, and contemporary issues* (pp. 925–944). Mahwah, NJ: Erlbaum.
- Friedman, M., & Rosenman, R. H. (1974). *Type A behavior and your heart*. New York: Knopf.
- Hart, P. M., & Cooper, C. L. (2002). Occupational stress: toward a more integrated framework. In N. Anderson, D. S. Ones, H. K. Sinangil, & C. Viswesvaran (Eds.), *Organizational psychology. Handbook of industrial, work and organizational psychology, Vol. 2* (pp. 93–114). Thousand Oaks, CA: Sage.
- Heitzmann, B., Schaarschmidt, U., & Kieschke, U. (2005). Diagnostik beruflichen Bewältigungsverhaltens bei Rehapatienten — die Leistungsmöglichkeiten des Verfahrens AVEM im Bereich der medizinischen Rehabilitation. [Coping patterns of patients in rehab — the achievement possibilities of the questionnaire AVEM at the medical rehabilitation]. *Praxis Klinische Verhaltensmedizin und Rehabilitation*, 70, 269–280.
- Hillert, A., & Schmitz, E. (Eds.). (2004). *Psychosomatische Erkrankungen bei Lehrerinnen und Lehrern*. [Psychosomatic diseases among teachers]. Stuttgart, Germany: Schattauer.
- Houston, B. K., Chesney, M. A., Black, G. W., Cates, D. S., & Hecker, M. H. L. (1992). Behavioral clusters and coronary heart disease risk. *Psychosomatic Medicine*, 54, 447–461.
- Jepson, E., & Forrest, S. (2006). Individual contributory factors in teacher stress: the role of achievement striving and occupational commitment. *British Journal of Educational Psychology*, 76, 183–197.
- Kieschke, U. (2003). *Arbeit, Persönlichkeit und Gesundheit [Work, personality, and health]*. Berlin: Logos.
- Kieschke, U., & Schaarschmidt, U. (2003). Bewältigungsverhalten als eignungsrelevantes Merkmal bei Existenzgründern: Ergebnisse einer Längsschnittstudie. [Coping style as indicator of entrepreneurs fitness: results of a longitudinal project]. *Zeitschrift für Personalpsychologie*, 2, 1–11.
- Kieschke, U., & Schaarschmidt, U. (2005). Bewältigungsverhalten bei Existenzgründern — Ein typendiagnostischer Beitrag zur Psychologie des Unternehmertums. [Coping patterns of entrepreneurs: a type-diagnostic approach]. *Wirtschaftspsychologie*, 2, 71–85.
- Klusmann, U., Kunter, M., Trautwein, U., & Baumert, J. (2006). Lehrbelastung und Unterrichtsqualität aus der Perspektive von Lehrenden und Lernenden. [Teachers stress and the quality of instruction: linking teachers' and students' perception]. *Zeitschrift für Pädagogische Psychologie*, 20, 161–173.
- Kyriacou, C. (2001). Teacher stress: directions for future research. *Educational Review*, 53, 27–35.
- Lambert, R. G., & McCarthy, M. (Eds.). (2006). *Understanding teacher stress in an age of accountability*. Greenwich, CT: Information Age Publishing.
- Laux, A., Ksienzyk-Kreuziger, B., & Kieschke, U. (2007). Unterstützung von Führungsarbeit und Teamentwicklung an der Schule. [Support of management work and team development at school]. In U. Schaarschmidt, & U. Kieschke (Eds.), *Gerüstet für den Schulalltag. Psychologische Unterstützungsangebote für Lehrerinnen und Lehrer* (pp. 93–116). Weinheim, Germany: Beltz.
- Lent, W. L., Brown, S. D., & Hackett, G. (1994). Toward a unifying social cognitive theory of career and academic interest, choice, and performance. *Journal of Vocational Behavior*, 45, 79–122.
- Maslach, C. (1982). Burnout: a social psychological analysis. In J. W. Jones (Ed.), *The burnout syndrome: Current research, theory, interventions* (pp. 30–53). Parke Ridge, UK: London House Press.
- Montgomery, C., & Rupp, A. A. (2005). A meta-analysis for exploring the diverse causes and effects of stress in teachers. *Canadian Journal of Education*, 28, 458–486.

- Prenzel, M., Baumert, J., Blum, W., Lehmann, R., Leutner, D., Neubrand, M., et al. (Eds.). (2005). *PISA 2003. Der zweite Vergleich der Länder in Deutschland: Was wissen und können Jugendliche?*. [PISA 2003. The second comparison between the German states: What do students know and can do?] Münster, Germany: Waxmann.
- Richardson, P. W., & Watt, H. M. G. (2005). "I've decided to become a teacher": influences on career change. *Teacher and Teacher Education*, 21(5), 475–489.
- Richardson, P. W., & Watt, H. M. G. (2006). Who chooses teaching and why? Profiling characteristics and motivations across three Australian universities. *Asia-Pacific Journal of Teacher Education*, 34(1), 27–56.
- Schaarschmidt, U. (Ed.). (2005). *Halbtagsjobber? Psychische Gesundheit im Lehrerberuf*. [Part-time-jobber? Mental health in the teaching occupation]. Weinheim, Germany: Beltz.
- Schaarschmidt, U., & Fischer, A. W. (2001). *Bewältigungsmuster im Beruf. Persönlichkeitsunterschiede in der Auseinandersetzung mit den Arbeitsbelastungen* [Coping patterns in the context of work. Individual differences in dealing with occupational stress]. Göttingen, Germany: Vandenhoeck & Ruprecht.
- Schaarschmidt, U., & Fischer, A. W. (2003). *AVEM – Arbeitsbezogenes Verhaltens- und Erlebensmuster* [AVEM – Patterns of work-related coping behavior] (2nd ed.). Frankfurt/M, Germany: Swets & Zeitlinger.
- Schaarschmidt, U., & Kieschke, U. (Eds.). (2007). *Gerüstet für den Schulalltag. Psychologische Unterstützungsangebote für Lehrerinnen und Lehrer*. [Ready for school life. Psychological supporting proposals for teachers]. Weinheim, Germany: Beltz.
- Siegrist, J. (1991). Contributions of sociology to the prediction of heart disease and their implications for public health. *European Journal of Public Health*, 1, 10–21.
- Vandenberghe, R., & Huberman, A. M. (Eds.). (1999). *Understanding and preventing teacher burnout. A sourcebook of international research and practice*. Cambridge, UK: Cambridge University Press.
- Watt, H. M. G., & Richardson, P. W. (2007). Motivational factors influencing teaching as a career choice: development and validation of the FIT-Choice scale. *Journal of Experimental Education*, 75(3), 167–202.
- Weber, A. (2003). Frühpension statt Prävention? Zur Problematik der Frühinvalidität im Schuldienst. [Early pension instead of prevention? About the problem of early inability to work in the teaching profession]. *Arbeitsmedizin, Sozialmedizin, Umweltmedizin*, 38, 376–384.
- Weber, A. (2004). Krankheitsbedingte Frühpensionierungen von Lehrkräften. [Early retirement of teachers as a result of health problems]. In A. Hillert, & E. Schmitz (Eds.), *Psychosomatische Erkrankungen bei Lehrerinnen und Lehrern* (pp. 23–38). Stuttgart, Germany: Schattauer.